

Heart Physiology

1- Increasing preload of the cardiac muscle will

- ✓ Increase the ventricular wall tension
- Decrease the initial velocity of shortening
- Reduce the ventricular end-diastolic pressure
- Reduce the peak tension of the muscle

منبع:

Guyton physiology p:118

2- Stimulation of the sympathetic nerves of the heart

- Decreases the permeability of the fiber membrane to chloride and potassium ions
- Increases the permeability of the fiber membrane to sodium and potassium ions
- ✓ Increases the permeability of the fiber membrane to sodium and calcium ions
- Decreases the permeability of the fiber membrane to calcium and potassium ions

منبع:

Guyton physiology p:128

3- Vagal stimulation causes all the following in the cardiac cells except

- ✓ Depolarizes the cell membrane of the nodal fibers
- Makes this excitable tissue much less excitable
- Increases the permeability of the fiber membranes to potassium ions
- Makes the "resting" membrane potential of the sinus node fibers more negative

منبع:***Guyton physiology p:128*****4- Which of the following is identical both in skeletal and cardiac muscle cells**

- Action potential duration
- Contraction duration
- Ca ion source for contraction
- ✓ Ca ion role in contractility

منبع:***Guyton physiology p:112*****5- The greatest oxygen consumption is normally seen during**

- Rapid filling
- ✓ Isovolumic contraction
- Reduced filling
- Isovolumic relaxation

منبع:***Guyton physiology p:119***

6- Left ventricular work is normally higher than the work of the right ventricle because

The left ventricular wall is thicker than the right

The stroke volume of the left ventricle is higher than the right

The afterload of the left ventricle is higher than the right

The preload of the left ventricle is higher than the right

منبع:

Guyton physiology p:117

7- Most of the chemical energy that provides the work of contraction comes from

Lactate



Fatty acids

Proteins

Glucose

منبع:

Guyton physiology p:118

8- ST segment of the ECG coincides with which part of the ventricular action potential?

Resting phase

Plateau phase



Depolarization phase

Hyperpolarization phase

منبع:

Guyton physiology p:132

9- Sympathetic stimulation displaces the workoutput curve of the heart to

- ✓ Left and upward
- Right and downward
- Right and upward
- Left and downward

منبع:***Guyton physiology p:118*****10- The relative refractory period is**

- The period between atrial depolarization and activation of the His-Purkinje system.
- The period that the myocardial cell is unable to generate an action potential.
- The period from the beginning of the P wave to the beginning of the R wave.
- ✓ The period that the myocardial cell is difficult to generate an action potential.

منبع:***Guyton physiology p:111*****11- Which of the following best describes the isovolumic contraction of the left ventricle?**

- Initiates from aortic valve closure till it's opening
- Initiates from mitral valve closure till it's opening
- Initiates from aortic valve closure till mitral valve opening
- ✓ Initiates from mitral valve closure till aortic valve opening

منبع:***Guyton physiology p:115***

12- Stokes-Adams syndrome is due to

Ventricular failure

AV bundle "overdriven" by the rapid sinus impulses

- ✓ A-V bundle block and Purkinje fibers defect to emit its intrinsic rhythmic impulses
- Sinoatrial node arrest

منبع:

Guyton physiology p:127

13- In the atrial pressure curve, the v wave occurs

Due to slight backflow of blood into the atria

- ✓ Toward the end of ventricular contraction
- When the ventricles begin to contract
- During atrial contraction

منبع:

Guyton physiology p:114

14- The incisura in the aortic pressure curve signifies

The late diastole

The late systole

- ✓ The early diastole
- The early systole

منبع:

Guyton physiology p:114

15- Left ventricular pressure is the least in which of the following ECG parts

- R-T
- ✓ T-P
- Q-T
- Q-R

منبع:

Guyton physiology p:132

16- The resting membrane potential and the threshold in sinoatrial node equals to

- 90 mv and -65 mv respectively
- ✓ -55 mv and -40 mv respectively
- 90 mv and -45 mv respectively
- 55 mv and -20 mv respectively

منبع:

Guyton physiology p:124

17- The second heart sound can best be heard during which phase of the cardiac cycle?

- Ventricular rapid ejection
- End of ventricular diastole
- ✓ End of ventricular systole
- Ventricular rapid filling

منبع:

Guyton physiology p:114

18- The slow conduction in the A-V bundle is caused mainly by

- ✓ Diminished numbers of gap junctions
- Great conductance
- Increased fiber numbers
- Increased p cells

منبع:***Guyton physiology p:125*****19- In which of the following conditions, the cardiac axis may deviate to the right?**

- At the end of deep expiration
- ✓ At the end of deep inspiration
- In obese people
- When a person lies down

منبع:***Guyton physiology p:145*****20- The fastest transmission velocity of the action potential in the heart is seen**

- In the ventricular muscle
- ✓ In the purkinje fibers
- In the atrial muscle
- In the AV node

منبع:***Guyton physiology p:126***

21- During the ventricular repolarization, which area repolarizes first

The base of the ventricles

Septum and endocardial areas



The inner apical surface

The outer apical surface

منبع:

Guyton physiology p:143

22- In phase 3 of the cardiac action potential

Potassium ion permeability decreases and calcium ion permeability increases.

Calcium channels open and slow potassium channels close.

Both calcium and slow potassium channels open.



Calcium channels close and slow potassium channels open.

منبع:

Guyton physiology p:111

Blood Physiology

23- What is the main importance of "Walling off " effect?

To accelerate chemicals release



To delay the spread of bacteria

To faster the inflammatory responses

To decrease cytokines secretion

24- Where is the most iron found in the body?

- ✓ Hemoglobin
- Ferritin
- Myoglobin
- Transferrin

25- Which type of anemia is from hemolytic one?

- Blood loss anemia
- Aplastic anemia
- Megaloblastic anemia
- ✓ Embryonic erythroblastosis

26- Immediately after a blood vessel is ruptured, trauma to the vessel wall causes

- ✓ Vascular constriction
- Extrinsic pathway of blood coagulation activation
- Platelet plug formation
- Intrinsic pathway of blood coagulation activation

27- Which organ produces the highest number of RBCs during the middle trimester of gestation?

- ✓ Liver
- Bone marrow
- Lymph node
- Spleen

28- Where is fibrin stabilizing factor secreted from?

RBCs

Injured vascular wall and damaged tissues

WBCs

✓ Platelets

29- Which type of WBCs exhibit stronger chemotaxis?

✓ Neutrophils

Lymphocytes

Eosinophils

Basophils

30- Which factor is not involved in the blood clotting process?

✓ VI

III

X

I

31- Which clotting factor initiates the intrinsic pathway?

X

✓ XII

XI

III

32- The longest life span of white blood cells is for

- neutrophils
- ✓ macrophages
- monocytes
- basophils

33- Which of the following is the most essential regulator of RBCs production?

- ✓ Tissue oxygenation
- The number of RBCs
- Viscosity
- Anemia

34- Which mediators are involved in the second line of defense against infection?

- GM-CSF and G-CSF
- ✓ Selectin and ICAM-1
- Living bacteria and cellular exudate
- Histamine and bradykinin

Circulatory Physiology

35- In the circulation control which sentence is wrong?

- ☐ In local blood flow, growth factor have an important role in short term control
- ☒ in short term control lack of oxygen cause proliferation of micro vasculature
- ☐ Adenosine is one of the most powerful vasodilators.
- ☐ Control of tissue blood flow is based on minimum need of tissue

36- Which of the following reduces right atrium pressure?

- ☒ Heart failure
- ☐ Arteriole relaxation
- ☐ Increase in blood volume
- ☐ Increase in PRU

37- In the physics of circulation:

- ☐ Static pressure stands for diastolic pressure
- ☐ Laminar flow is appears when diameter increase
- ☐ Pulse pressure is the sum of sis+ Dias divided to 2
- ☒ Dynamic pressure acting on static pressure

38- Which of the following organs have highest and lowest blood flow per 100 gram of tissue respectively?

- ☐ Liver – muscle
- ☒ kidney - bone
- ☐ Heart- pulmonary system
- ☐ Thyroid-brain

39- Active hyperemia happens in:

- increase in blood pressure
- ✓ increase in tissue metabolism
- injury of tissue
- short term occlusion of artery

40- In the neural control of circulation;

- sympathetic system is the main part of vasomotor which activates by c center
- capillary have highest sympathetic innervation
- ✓ A1 system in vasomotor activates para sympatric system.
- para sympatric have main role in in blood vessels motion(vasomotion)

41- In a constant resistant blood vessel, which of the following has higher blood flow?

- Pressure in the beginning 300 and in the end of vessel is 260 (mm/Hg)
- Pressure in the beginning 250 and in the end of vessel is 200 (mm/Hg)
- Pressure in the beginning 100 and in the end of vessel is 40 (mm/Hg)
- ✓ Pressure in the beginning 80 and in the end of vessel is 10 (mm/Hg)

42- Which of the followings is not correct.

- Activation of chemo receptors Cause reduction in blood pressure .
- Baroreceptors have an important role in long term regulation of blood pressure
- ✓ In the bodies different positions, baroreceptors regulates blood pressure
- Atrial and pulmonary reflexes are sensitive to blood volume and regulates arterial pressure

43- In the flow issue, which one is correct?

Increase in length of vessels increases flow in it.

It has direct correlation with second power of radius.



Increase in pressure has direct effect on velocity and flow.

Increase in Viscosity has equal effect on increase of velocity and flow.

44- In which of the following vessels turbulent flow never does not appears?

Aorta artery

Pulmonary artery

Brachial artery



Arterioles

45- Which of the following sentences is not correct?

Amputation causes increase in PRU

Inhibition of sympathetic system increase critical closure point

In a parallel system increase in tubules decreases resistance

By developing the new vessels blood flow will increase.

46- In the normal heart working, which of the following is mostly used for producing energy?

Glucose

Galactose



Fatty acids

Adenosine

47- In the fluids physics which one is correct?

Blood viscosity is 1.5 centi poise

RBC is responsible for blood viscosity.

Proteins does not have any effect on osmotic pressure.



RBC Have eminent effect on osmotic pressure also.

48- Increases in which of the following Causes reduction in turbulent flow?

Diameter

Velocity

Density



Viscosity

49- Which of the followings is not duty of circulation?

Scavenger duty of the tissues environment

Mobilization of immune system

Temperature regulation and heat distribution



ATP production

50- In the series circuits which one is not correct?

They have variable flow

Disorder in any part collapses the circuits.

They have variable pressure

Total resistance is the sum of the resistances.

51- Which of the following tissue has the least sympathetic innervation?

- ✓ Brain
- Kidney
- Skin
- Intestine

52- In the capillary filtration which sentence is correct?

- ✓ Activation of pre- capillary sphincter is responsible for capillary vasomotion
- Protein content of lymph is equal with the plasma level of protein
- Activation of lymphatic pump increases local edema.
- Proteins easily passes through to inter cellular space via capillary fenestra

53- In the blood pressure which one is correct?

- In beginning of iso-volume contraction we can hear second sound of heart.
- High diastolic pressure is a good sign of blood vessel healthiness.
- In the vessels filling pressure reduces by vaso constriction.
- ✓ Diastolic pressure is a pressure which is in the arteries in the ventricle diastolic time.

54- Which one increases pulse pressure?

- Increase in arterial compliance
- Decrease in stroke volume
- Increase in end diastolic volume
- ✓ Increase in cardiac output

55- Which of the following is not involved with histamine effect?

- Allergy
- Edema
- ✓ Arteries contraction
- Tissue injury

56- Which of the followings doesn't help to venues return to heart?

- Capillary blood pressure
- Activity of adjacent muscles
- ✓ Hydrostatic pressure
- Activity of veins valves

57- Which of the followings causes an increase in the PRU?

- Para sympatic tone
- Organ transplantation
- Bigger radius
- ✓ Poly- cytemia

58- Which of the following has the strongest impact in regulation of blood flow?

- ✓ Radius
- Blood Pressure changes
- Viscosity
- Arteries length

Anatomy

59- The latissimus dorsi muscle is involved in all of the following actions except..

- Adduction
- Extension
- ✓ Flexion
- Medial rotation

60-

At the opening of all of the following vessels you can find valve except.....

- Coronary sinus
- Vena cava inferior
- Pulmonary trunk
- ✓ Right pulmonary vein

61- The anterior surface of the aortic arch is adjacent to which of the following structures?

Left laryngeal ligament

Deep cardiac plexus

Right phrenic nerve

✓ Left vagus nerve

منبع:

Grays for student Anatomy

62- All the following elements are observed in the right atrium except ...?

Fossa ovalis

✓ Infundibulum

Pectinate muscles

Orifice of coronary sinus

63-

Which of the following structure is felt back of the left atrium?

Right pulmonary veins

Right atrium

✓ Esophagus

Right pulmonary veins

64-

The lymph of all the following areas is drained to the thoracic duct except ...?

- ✓ Left upper limb
- Right lower limbs
- Left lower limbs
- Right upper limb

65- All of the following joints are type of the synovial joint , except?

- Third costo sternal
- Costo vertebral
- ✓ Sternal angle
- Costo transverse

66- All the following elements are present in the posterior mediastinum space except.....?

- Azygos vein
- Thoracic duct
- ✓ Left brachiocephalic vein
- Thoracic aorta

67- Phrenic nerve innervate all of the following structures except...

Diaphragm

Parietal layer of serous pericardium

Fibrous pericardium

✓ Visceral layer of serous pericardium

68-

All of the bones participated in forming the superior aperture of thorax except..

First thoracic vertebrae

Sternum

✓ Second thoracic vertebra

First rib

69-

Which of the following veins is passing through anterior interventricular sulcus?

Median cardiac vein

Coronary sinus

✓ Greater cardiac vein

Lesser cardiac vein

70- In which of the following ribs tubercle is absent?

- ☐ 7th
- ☒ 12th
- ☐ 1th
- ☐ 2th

منبع:

Grays for student Anatomy

71- Head of the third rib is going to articulate with which thoracic vertebrae

- ☒ Body of T2 and T3
- ☐ Body of T4
- ☐ Body of T3 and T4
- ☐ Body of T3

منبع:

Grays for student Anatomy

72- About right vagus nerve all of the following statements are correct except....?

- ☒ It is passing anterior of vena cava superior
- ☐ It is passing through esophageal hiatus of diaphragm
- ☐ It is passing posterior to arch of aorta
- ☐ It is passing posterior of esophagus

73- Which of the following arteries are going to find in posterior ventricular sulcus in right dominate heart

- ✓ A branch from right coronary artery
- Right marginal artery
- Left marginal artery
- Circumflex

74-

The right 10th posterior intercostal vein is drained to which of the following veins

- Accessory hemiazygos vein
- Vena cava superior
- Hemiazygos vein
- ✓ Azygos vein

75-

-All the following muscles are attached to the ribs except..

- ✓ Trapezius
- Serratus posterior superior
- Serratus posterior inferior
- Serratus anterior

76- All the following arteries arise from the internal thoracic artery except....

Musculocutaneous artery

Anterior fifth intercostal artery

Superior epigastric artery

✓ Posterior first intercostal artery

77- Lateral costo transvers ligament attached to which of the following structures?

Neck of the rib

Body of thoracic vertebrae

Spinous process

✓ Tubercle of the rib

78- Which of them arises from thoracic aorta?

Anterior 5th intercostal artery

Pericardiophrenic artery

✓ Subcostal artery

Anterior 4th intercostal artery

79- All of the following statments about symphatic trunk are true except...

White rami communicantes is seen in the thorax

Superior part of sympathetic trunk is located in the neck

✓ The 11th sympathetic ganglion is involved in splanchnic major

Gray rami communicantes is seen in the thorax

80- All of the followings nerves have a lateral cutaneous branch except...

6th intercostal nerve

✓ First intercostal nerve

Second intercostal nerve

3th intercostal nerve

81- When people are standing, the base of the heart is in front of which of the thoracic vertebrae?

T5-T8

T3-T4

✓ T6-T9

T10-T12

82- Chorda tendino in heart are going to attached to which of the following valves?

- ✓ Tricuspid valves
- Right Pulmonary valve
- Left Pulmonary valve
- Aortic valve

83-

Which of the following elements belongs to the shadow of the right border of the heart in a radiographic image?

- Arch of aorta
- Right and left ventricles
- Pulmonary trunk
- ✓ Right atrium

Histology

84-

Which cell produce antibodies?

- macrophage
- ✓ Plasmacell
- B cell
- T cell



85-

Which antibody has dimer structure?

IgM

IgE

IgG

☒ IgA

86-

The thinnest layer of vessel walls is?☒ Intima

Adventitia

media

seroza

87- where is the site of blood filtration?☒ spleen

thymus

lymphnode

tonsil

88- Which antibody can cross placenta?

- ☐ IgA
- ☒ IgG
- ☐ IgM
- ☐ IgE

89-

In thymus where we can find Hassall's body ?

- ☐ Cortex
- ☒ medulla
- ☐ Capsule
- ☐ trabecula

90-

What is the other name of adenoid?

- ☐ Lingual tonsil
- ☐ Palatine tonsil
- ☒ Pharyngeal tonsil
- ☐ Cervical lymph node

91- The thickest layer in the wall of heart is?

- endocardium
- ✓ myocardium
- pericardium
- epicardium

92-

Which one is not secondary lymphoid organ?

- tonsil
- MALT
- spleen
- ✓ thymus

93- Where are T lymphocytes in lymphnodes?

- Cortex
- medulla
- ✓ paracortex
- Cortical sinvs

Embryology

94- Which mesodermal layer gives rise to the endocardial heart tubes?

- ✓ Splanchnic mesoderm
- Paraxial mesoderm
- Somitc mesoderm
- Somatic mesoderm

95- The embryo cardinal venous vessels which drain into the sinus venosus are:

- superior cardinal veins
- inferior cardinal veins
- anterior cardinal veins
- ✓ common cardinal veins

96- A newborn infant presents with cyanotic heart disease and fails to thrive. On examination of the heart, it was revealed that the aorta arose from the right ventricle and the pulmonary trunk arose from the left ventricle. What is the name given to this type of cardiac malformation?

- Ventricular septal defect (VSD)
- Tetralogy of Fallot
- ✓ Transposition of the great vessels
- Persistent truncus arteriosus

97- The left sinus horn becomes the

Inferior vena cava.

Main inflow of the right atrium

✓ Coronary sinus.

Oblique vein of the right atrium.

98- Which heart chamber does the right horn of the sinus venosus contribute to during development of the fetal heart?

Left atrium

Left ventricle

Right ventricle

✓ Right atrium

99- The sinus venarum is

Formed from the right auricle.

Formed from the pulmonary veins.

The trabeculated part of the right atrium.

✓ Formed from tissue of the right sinus horn

100- Which of the following statements concerning the aorticopulmonary septum is INCORRECT?

The aorticopulmonary septum partitions the outflow channel of the ventricles into the aorta and the pulmonary trunk.

The aorticopulmonary septum undergoes a 180° spiral

✓ The aorticopulmonary septum contributes to the formation of the muscular interventricular septum

The aorticopulmonary septum is largely derived from neural crest cells